

WASP-3b

R-1044

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_160522 · created 2026-07-30T08:42:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457530.9432 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.036
Transit depth (Rp/Rs)^2	1.5 +/- 0.88 [%]
Orbital Inclination (inc)	79.42 +/- 3.05
Airmass coefficient 1 (a1)	1.508 +/- 0.029
Airmass coefficient 2 (a2)	-0.392 +/- 0.077
Scatter in the residuals of the lightcurve fit is	1.91 %
Best Comparison Star	None
Optimal Aperture	2.86
Optimal Annulus	4.94
Transit Duration (day)	0.058 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.036 vs 0.10538 (deviation 0.46σ)
Duration fit vs published	0.058 d vs 0.1167 d (deviation 1.73σ)
Mid-time O–C	-5.2 min
Depth SNR	3.4
Residual scatter	1.91 %

Light curve

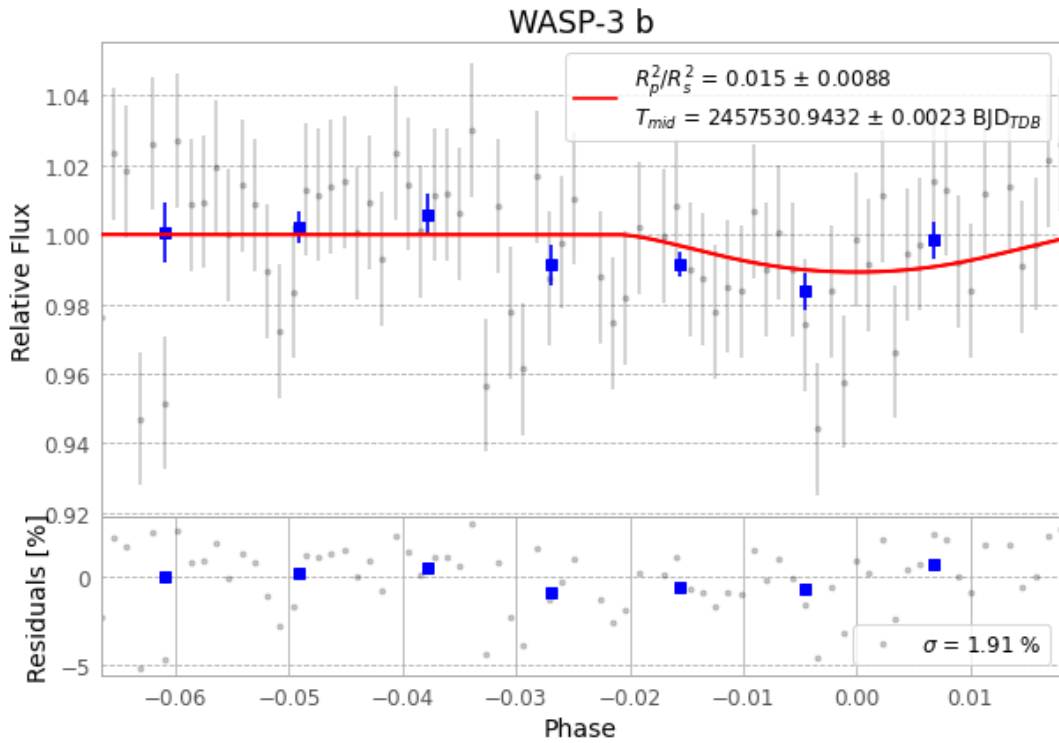


Figure 1 — FinalLightCurve_WASP-3 b_2016-05-22.png (SHA-256 1c2228c6f6b58fb3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 515d73bb8c85cc6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-3160522073612.FITS → WASP-3160522112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-160522.log (59a13956063a...), FinalLightCurve_WASP-3 b_2016-05-22.png (1c2228c6f6b5...), FinalLightCurve_WASP-3 b_2016-05-22.csv (826b41dd0ea2...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 08:42Z.